

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester III Examination, 2023-24

Paper: GLB-301 — Petrology and Economic Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 067528

*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory.*

SECTION — A

Question 1. Write a brief note on the phase rule application to the H₂O system with a suitable diagram. [14]

Question 2. Discuss the classification of sedimentary rocks. [14]

Question 3. Briefly describe metamorphic textures and structures. [14]

Question 4. Write short notes on **any four** of the following: $[4 \times 3\frac{1}{2} = 14]$

- (a) Bowen's reaction series
- (b) Batholith
- (c) Conglomerate
- (d) Shale
- (e) Progressive metamorphism
- (f) Hornfels

SECTION — B

Question 5. Write important chemical formula, distribution in India and uses of **any four** of the following economic minerals: $[4 \times 3\frac{1}{2} = 14]$

- (a) Halite
- (b) Quartz
- (c) Biotite
- (d) Pyrite
- (e) Malachite
- (f) Monazite

Question 6. Write in brief **any two** of the following: [2 × 7 = 14]

- (a) What is petroleum? Discuss the origin and economic uses of petroleum.
- (b) What are fertilizer minerals? Discuss the economic uses of fertilizer minerals and their distribution in India.
- (c) What are galena and sphalerite? Describe the economic uses and distribution of lead-zinc ore deposits in India.

Question 7. Briefly describe the physical properties of manganese ore minerals, economic uses, and distribution of mineral deposits in the map of India. [14]

SECTION — C

Question 8. Answer the following:

- (a) What is the mineralogical composition of peridotite? [1]
- (b) Differentiate between breccia and conglomerate. [2]
- (c) Define the matrix of sedimentary rocks. [2]
- (d) What is slate? [1]
- (e) Write the name of a low-grade metamorphic rock. [1]
- (f) Write the names of two coal deposits in India. [2]
- (g) Write the name of an ore mineral of beryllium. [1]
- (h) How many sets of cleavage planes are found in kyanite? [1]
- (i) Which metal is extracted from magnesite? [1]
- (j) What are the economic uses of chalcopyrite? [2]